Project Title

Diabetes Risk Prediction Using Logistic Regression and Random Forest with Cross-Validation

Project Report submitted for the partial fulfilment of grade for the course

MACHINE LEARNING(25SC2107E)

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ABSTRACT

Academic and intellectual integrity relies on the effective identification of copied content across large document collections. This project presents a robust plagiarism detection system developed specifically to identify duplicate passages using efficient rolling hash techniques. Grounded in the principles of the Rabin-Karp string-matching algorithm, the application generates polynomial hash values for sliding windows of text, enabling rapid identification of matching segments across multiple documents. To ensure maximum accuracy, the system implements rigorous collision handling mechanisms, such as double hashing, preventing false positives when different text passages produce identical hash values. Adhering strictly to core system constraints, all underlying hash structures, string processing algorithms, and array manipulations are engineered entirely from scratch without relying on built-in standard library utilities like `java.util.*`. Beyond detecting matches, the engine calculates comprehensive similarity percentages to quantify document overlap accurately. Furthermore, it visually highlights duplicated passages within the interface to provide intuitive, actionable output for end users. Ultimately, this project demonstrates a high-performance, sub-quadratic approach to multi-document comparison and practical string analytics.
Predicting diabetes risk is a vital healthcare objective that requires a scalable, end-to-end machine learning lifecycle. This project presents an integrated predictive system that evaluates patient medical records using both linear and ensemble modeling paradigms. During the feature engineering phase, continuous clinical attributes are standardized, and Principal Component Analysis alongside K-means clustering is applied to discover latent patient risk groups, which serve as engineered features. We implement Logistic Regression using cross-entropy loss to establish a linear decision boundary baseline, comparing its performance against a Random Forest classifier that leverages bagging and feature subsampling to capture complex non-linear feature interactions.
To ensure unbiased evaluation and prevent data leakage, model tuning and selection strictly employ stratified k-fold cross-validation. Performance is comprehensively evaluated using ROC-AUC, Precision, Recall, and probability calibration. Finally, the optimized model artifact is packaged and served via a low-latency FastAPI REST endpoint equipped with inference monitoring hooks for real-time risk assessment. This work demonstrates a production-ready machine learning solution bridging data ingestion, feature engineering, statistical model selection, and deployment engineering.